GGG-S-280F
August 2, 1972
SUPERSEDING
Fed. Spec. GGG-S-280E
July 10, 1961

FEDERAL SPECIFICATION

SHEARS, GRASS CUTTING AND SHEARS, HEDGE TRIMMING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers grass shears and hedge shears of several types, classes, and styles.

1.1.1 Federal specification coverage. Federal specifications do not cover all types, classes, and styles of shears which are commercially available, but only those which are generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, styles, and classes. The shears covered by this specification shall be of the following types, styles, and classes, as specified (see 6.2).

Type I - Grass-cutting.
  Class 1 - Scissors action.
    Style A - Flat handgrip (see figure 1).
    Style B - Offset handgrip (see figure 2).
  Class 2 - Draw action (see figure 3).
  Class 3 - Sheep-shear type (see figure 4).

Type III - Hedge-trimming (see figure 5).
  Class 1 - Serrated blade(s).
  Class 2 - Plain blades.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as hereinafter specified and shall be of good quality throughout.

3.2 Illustrations. The illustrations shown herein are descriptive and not restrictive, and are not intended to preclude the purchase of shears which are otherwise in accordance with the requirements of this specification.

3.3 Blades. Blades shall be of good quality carbon steel, properly heat treated. The cutting edge of each blade shall be uniformly ground throughout its length and sharpened ready for service without need for further attention. The blades shall be pointed. The outer surfaces of the blades shall be flat, beveled, or oval shaped, at the option of the contractor.

3.4 Hardness of blades. Blades shall be heat treated to a hardness within the range of Rockwell C 55 to 61.

3.5 Toughness of blades. There shall be no cracking or chipping when any blade is subjected to the toughness test specified in 4.4.2.

3.6 Handgrips. Metal handgrips, where hereinafter specified, shall be free from burrs, fins, seams, rough edges, sharp corners and shall be capable of passing the drop test specified in 4.4.5. Wood handgrips, where hereinafter specified, shall be sound, well-seasoned hardwood. The blade ends of wood grips shall have substantial metal ferrules.

3.7 Finish.

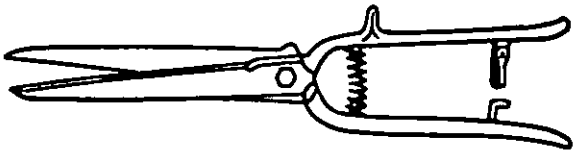
3.7.1 Blades. Each blade shall have the inner surface ground not coarser than that resulting from a 60-grit abrasive wheel or belt. All bare-metal surfaces, except blades, shall have a plated, painted, lacquered, or enameled finish, at the option of the contractor, except that oil coating is acceptable on type III shear blades, shanks, and tangs.

3.7.2 Handgrips. Handgrips, if made of wood, shall be smooth and shall have an applied coating of varnish, lacquer, or paint, free from tackiness.

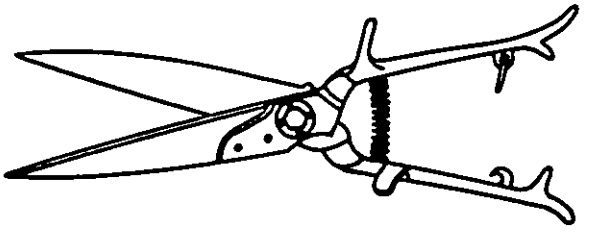
3.8 Identification marking. The shears shall be legibly and permanently marked with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.9 Type I, class 1 scissor-action grass-cutting shears. The type I, class 1 shears shall have a scissors-type action whereby both blades move equally toward (or from) a centerline which bisects the angle of the opposing blades. The shears shall be similar to figure 1 or figure 2 and comply with the dimensions in table I. The shears shall be equipped with an opening spring, stop(s), and locking device.

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Integral handgrip.



Nonintegral handgrip.

FIGURE 1. Type I, class 1, style A scissors type, flat handgrip grass cutting shears (integral or nonintegral handgrip optional).

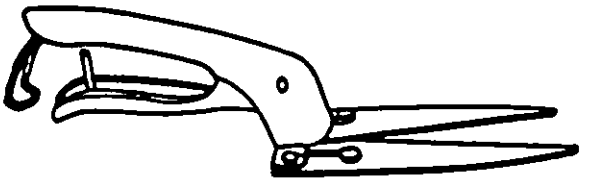


FIGURE 2. Type I, class 1, style B scissors type, offset handgrip, grass cutting shears.

TABLE I. Type I grass-cutting shears

Class	Style	Blade				Overall length 4/	
		Length of cut 1/		Width 2/	Thickness 3/		
		Minimum	Maximum	Minimum	Minimum	Minimum	Maximum
		Inches	Inches	Inches	Inch	Inches	Inches
I	A	4	6-1/4	1/2	0.060	11	13-1/8
I	B	4	6-1/4	1/2	.080	11	13-1/8
II		4-1/2	6-1/4	3/4	.085	12-3/4	14-3/4
III		4-3/4	6-1/8	1-5/16	.060	12	13-1/8

1/ Measured in a straight line along the cutting edge from the point to the end of the cut.

2/ Measured across the blade at the midpoint of the length of cut.

3/ Measured, exclusive of coating, at the thickest section at the midpoint of the length of cut.

4/ Measured in a straight line parallel to the plane of the blades from the point of the shears to the extreme end of the handgrip or bow, with the shears in closed position.

3.9.1 Joints of type I, class 1 shears.

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3.9.1.1 Type I, class 1, style A. The blades of flat handgrip style shears shall be joined by means of a steel bolt or screw, not less than 1/4-inch diameter, in a manner such that there is no excessive looseness, sidewise movement, or binding when shears are opened or closed. The bolt or screw shall be threaded in one blade and snugly fitted in the other blade. The bolt or screw may be with or without a nut. The bolt or screw, or the nut, shall be locked in a manner such that it will not work loose when shears are in use. If a bolt is used, the head of the bolt shall be seated on a steel spring tension washer.

3.9.1.2 Type I, class 1, style B. The blades of offset handgrip style shears shall operate in a manner such that there is no excessive looseness, sidewise movement, or binding when shears are opened or closed during the cutting operation.

3.9.2 Handgrips.

3.9.2.1 Type I, class 1, style A (figure 1). Flat handgrip style shears shall be arranged so the plane of operation of the handgrips is parallel to the cutting plane of the blades. Unless otherwise specified (see 6.2), the handgrips shall be either integral or nonintegral. Each handgrip shall be not less than 1/2 inch thick and not less than 1/4 inch wide at a point 3 inches from the free end of equivalent cross-sectional area. Nonintegral handgrips, if riveted, shall be malleable iron or steel forgings and one end of each handgrip shall be fitted to the blade and secured either with at least two rivets not less than 0.094-inch diameter or other equally strong connection. Rivet heads, if any, shall be flush with the inside surface of the blade and shall be neatly headed on the outside.

3.9.2.2 Type I, class 1, style B (figure 2). Offset handgrip style shears shall be arranged so the plane of operation of the handgrips is at right angles to the cutting plane of the blades. Handgrips shall be not less than 0.060 inch thick metal suitable for the purpose intended. The handgrips shall be joined by means of either a threaded steel bolt and nut not less than 3/16 inch in diameter or a rivet of not less than 3/16 inch diameter.

3.9.3 Handgrip clearance. The handgrip clearance at the extreme end of the handgrip of type I, class 1, style A shears shall be not less than 1-1/2 inches, determined in the manner specified in 4.4.3. Type I, class 1, style B shears shall have a clearance of at least 1/4 inch.

3.9.4 Spring. The spring shall be made of hard-drawn or oil-tempered spring-steel wire and shall be of substantial construction and suitable for heavy-duty service. The spring shall be assembled in a manner such that it can be readily replaced. Wire for springs shall be not less than 0.050 inch diameter.

3.9.5 Locking device. The locking device shall be of substantial construction, and shall be arranged to lock the blade in closed position when the shears are not in use.

3.9.6 Stops. The closing of the shears shall be limited by means of either integral or permanently fastened stop(s).

3.10 Type I, class 2, draw type, grass cutting shears. The shears shall comply with the requirements of table I and shall be similar to figure 3. The shears shall be equipped with an opening spring, positive stop(s), and locking device. The action shall be by bar or rod pulling or pushing the top blade across the stationary lower blade. A pressure plate or similar device shall be used to keep the blade from separating under heavy cutting pressure.

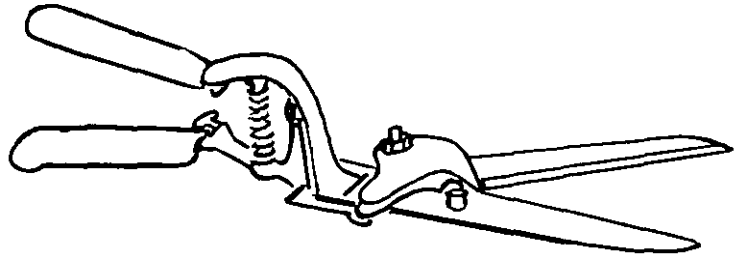


FIGURE 3. Type I, class 2, draw type, grass cutting shears.

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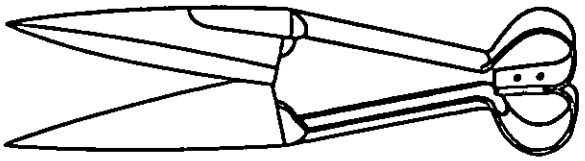
3.10.1 Handles. The handles shall operate in an up and down action at right angles to the cutting plane of the blades. The handgrips shall be cast iron, aluminum die casting, or drop forged. The handgrip portion of the handles shall be vinyl covered, chrome plated, or other non-corrosive finish.

3.10.2 Spring. The spring shall comply with the spring requirements for type I shears, as specified in 3.9.4.

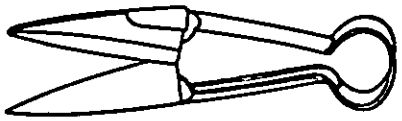
3.10.3 Locking device. The locking device shall comply with the locking device requirements for type I shears, as specified in 3.9.5.

3.10.4 Stops. The stops shall comply with the stop requirements for type I shears, as specified in 3.9.6.

3.11 Type I, class 3, sheep-shear grass cutting shears (figure 4). The shears shall comply with the requirements of table I. Unless otherwise specified in the invitation for bids (see 6.2), the shears shall be either double or single bow.



Double bow



Single bow

FIGURE 4. Type I, class 3, sheep-shear-type, grass cutting shears.

3.11.1 Fabrication. Double-bow shears shall have an integrally-formed bow blade with the bows securely fastened together with two or more rivets. Single-bow shears shall have the blades, handgrips, and bow formed from one piece of sheet steel without any riveting or welding of members. The cross-sectional dimensions, determined at the midpoint of the back of each bow of double-bow handgrips and at the midpoint of the back of the bow of single-bow handgrips shall be not less than 0.068 inch (thickness) by not less than 1-5/8 inches (width) or of equivalent cross-sectional area.

3.11.2 Bows. Bows shall have proper spring qualities so that the shears are always in the proper open position for cutting and are returned to the open position upon release of the handgrips after the cut is made. When the shears are in the open position, the heels of the blades shall have an overlap of at least 1/4 inch.

3.11.3 Handgrip clearance. The handgrip clearance at the end of the shears shall be not less than 3/4 inch, determined in the manner specified in 4.4.3.

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3.12 Type III, hedge-trimming shears (figure 5). The shears shall comply with the requirements of table II for size stated in the invitation for bids.

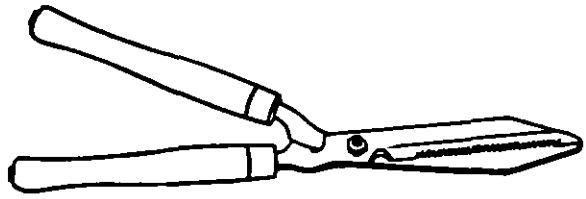


FIGURE 5. Type III, hedge trimming shears.

Class 1 - Serrated blade(s).

Class 2 - Plain blades (not shown).

TABLE II. Type III, classes 1 and 2 hedge shears

Size[5]	Blade				Overall length[4]	
	Length of cut[1]		Width[2]	Thickness[3]		
	Minimum	Maximum	Minimum	Minimum	Minimum	Maximum
Inches	Inches	Inches	Inches	Inch	Inches	Inches
8	7-1/2	9-1/8	1-1/8	0.170	20	24
9-1/2	8-7/8	10-1/4	1-5/16	.170	21	27

1/ Measured in a straight line along the cutting edge from the point to the end of cut.

2/ Measured across the blade at the midpoint of the length of cut.

3/ Measured, exclusive of coating, at the thickest section at the midpoint of the length of cut.

4/ Measured in a straight line parallel to the plane of the blades from the point of the shears to the free end of the handgrips with the shears in closed position.

5/ The nominal size is determined by the length of the blade measured from the center of the joint to the end of the blade.

3.12.1 Blades. Type III, class 1 shears shall have either one plain blade and one serrated blade or two serrated blades. Type III, class 2 shears shall have two plain blades. When specified (see 6.2), type III shears shall have a cutting notch or notches adjacent to the joint, suitable in size, shape, and cutting ability to shear thick stems. Blades shall be integral with the shanks of the handgrips.

3.12.2 Joints. The shears shall be substantially joined by means of a steel bolt in a manner such that there is no excessive looseness, sidewise movement, or binding when shears are opened or closed. The bolt shall be not less than 3/8-inch diameter and shall have a threaded steel nut.

3.12.3 Handgrips. Unless otherwise specified (see 6.2), either wood handgrips or tubular-steel handgrips shall be furnished (see 3.11.3.1 and 3.11.3.2).

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3.12.3.1 Wood handgrips. The diameter of each wood handgrip, determined at the smallest cross section, shall be at least 1 inch. The overall length of the handgrip shall be not less than 9 inches. Handgrips shall be hardwood. Each handgrip shall be riveted, pinned, or bolted to the tang end of the shank, at the option of the contractor. If the handgrip is riveted or pinned, the tang shall extend at least 2-1/4 inches in the handgrip. If riveted, there shall be one or more rivets through both sides of the ferrule, handgrip, and tang. If pinned, there shall be one or more threaded steel drive pins through one side of the ferrule and handgrip, through the tang, and into the other side of the handgrip. Rivets and drive pins shall be neatly headed. If the handgrip is bolted, the tang end shall be threaded and the tang shall extend through the handgrip with the free end of the tang secured to the top of the handgrip by means of steel nut seated on a steel washer.

3.12.3.2 Tubular-steel handgrips. The tubular-steel handgrips shall be covered with rubber or rubber-like plastic cushion on the handgripping portion. The tubular steel shall be securely fastened to the tang on the blades. The rubber cushion shall adhere tightly to the tubular steel. The tubular steel shall be plated for resisting corrosion.

3.12.4 Handgrip clearance. The handgrip clearance at the end of the shears shall be not less than 2 inches, determined in the manner specified in 4.4.3.

3.12.5 Stops. The closing of the shears shall be limited by means of either integral or permanently attached stops on either the heels or shanks of the blades at the option of the contractor.

3.13 Workmanship. Workmanship shall be first class in every respect. The shears shall be free from defects and imperfections that might affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.3 Examination. Samples selected in accordance with 4.2 shall be examined for the following defects. The Acceptable Quality Level (AQL) shall be 2.5 percent defective.

1. Dimensions not as specified.
2. Cutting edges not as specified.
3. Tacky finish on wooden handgrips.
4. Moving parts do not function.
5. Assembly not complete.
6. Spring action does not open blades to full cutting position.

4.4 Tests. Samples selected in accordance with 4.2 shall be tested as follows. Failure to pass any test shall constitute a defective unit. AQL shall be 1.0 percent defective.

4.4.1 Hardness of blades. The hardness shall be determined for compliance with the requirements of 3.4. Hardness not within the specified limits shall constitute failure of this test.

4.4.2 Toughness of blades. Each test sample shall be subjected to the following test to determine compliance with the requirements of 3.5.

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4.4.2.1 The shears shall be opened, disassembled, or otherwise arranged in open position and one blade, with the cutting edge up, secured in a suitable support such as a vise. One end of a test rod 24 inches long and 1 inch in diameter shall be supported on a rest at approximately the same level as the cutting edge of the shears. (The test rod shall be made of cold-rolled carbon steel in an "as rolled" condition, approximately 150 Brinell hardness.) The other end of the rod shall be elevated to a height of 6 inches above the cutting edge and released. The rod shall strike the cutting edge at a distance of 2 inches from the elevated end and at right angles to the plane of the blade. Cracking or chipping of the blade shall constitute failure of this test.

4.4.3 Handgrip clearance. The handgrip clearance shall be determined for compliance with the requirements of 3.9.3, 3.11.3, and 3.12.4, as applicable. The blades in normal open position shall be placed on a plane surface, so as to rest parallel with the surface. The distance between the plane surface and the lowest portion of the extreme ends of the handgrips or bows shall be the amount of clearance. Handgrip clearance not within the measurements specified, shall constitute failure of this test.

4.4.4 Hardness of bows. The hardness of bows shall be in accordance with 3.11.2. Hardness not within the specified limits shall constitute failure of this test.

4.4.5 Drop test. Each sample shear with metal handgrips shall be dropped on a concrete floor from a height of 5 feet so that it strikes directly on each handgrip at least once. After impact, the sample shears shall show no evidence of cracks, fractures, breaks, or bends in the handgrips or components.

4.5 Inspection of preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be inspected in accordance with PPP-P-40 to verify compliance with the requirements in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. The grass and hedge shears covered by this specification are intended for use in the cutting of grass and trimming of hedges and plants.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style, as applicable (see 1.2.1).
(c) If a specific kind of bow is required, specify if double or single bow shears are to be furnished (see 3.11).
(d) If integral or nonintegral handgrips are required, so specify (see 3.9.2.1).
(e) If cutting notch(es) are required on type III shears, so state (see 3.12.1).
(f) If a specific kind of handgrip is required for type III hedge-trimming shears, state the kind required (see 3.12.3).
(g) State level of packaging and packing protection (see section 5).

6.3 Supersession data. Type II shears in GGG-S-280E are designated as type I, style C in this specification.

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